

Supplementary Material: Fixed-Spectrum Swaps Separate Attention Capacity from Assignment

Anonymous submission

This document provides proofs, experimental protocols, full result tables, secondary figures, negative replications, and reproducibility information for results summarized in the main paper.

Guide to Supplementary Evidence

Table S1 maps the main claims to their detailed evidence and archived analysis. The anonymous bundle also contains `ARTIFACT_INDEX.md`, which gives exact input and output paths, and `REPRODUCE.md`, which separates saved-artifact analysis from GPU reruns.

Theory Details

Established softmax grouping identity

For evidence positions S , distractors D , scores s_j , and temperature $T > 0$, let $Z_G = \sum_{j \in G} e^{s_j/T}$ and $F_G = -T \log Z_G$. Direct grouping of the softmax denominator gives

$$m_S := \sum_{j \in S} a_j = \frac{Z_S}{Z_S + Z_D} = \sigma\left(\frac{F_D - F_S}{T}\right). \quad (1)$$

Defining the group log-mean-exp score $\bar{s}_G = T \log(|G|^{-1} \sum_{j \in G} e^{s_j/T})$ yields the exact threshold relation

$$m_S \geq \alpha \iff \bar{s}_S - \bar{s}_D \geq T \left(\log \frac{|D|}{|S|} + \text{logit}(\alpha) \right). \quad (2)$$

These standard Gibbs or partition-function identities (Ramsauer et al. 2021) describe the competition between evidence and distractors. Equation 2 makes the dependence on distractor scores explicit: distractor count alone is insufficient because arbitrarily many negligible-score distractors need not reduce evidence mass.

Fixed-spectrum routing range

Let $a_{(1)} \geq \dots \geq a_{(n)}$, let S contain r evidence positions, and define

$$C_r^+ = \sum_{i=1}^r a_{(i)}, \quad C_r^- = \sum_{i=n-r+1}^n a_{(i)}, \quad (3)$$

$$K_r(a) = C_r^+ - C_r^-.$$

Theorem S1 (Capacity-assignment decomposition). *For every permutation π of a fixed attention-weight multiset, $C_r^- \leq m_S(\pi) \leq C_r^+$, both endpoints are attainable, and for $K_r(a) > 0$ the unique coordinate*

$$\rho_S(\pi) = \frac{m_S(\pi) - C_r^-}{K_r(a)} \in [0, 1] \quad (4)$$

gives $m_S(\pi) = C_r^- + \rho_S(\pi)K_r(a)$.

For affine margin $M(\pi) = b + \sum_j a_{\pi(j)}u_j$, sorted utilities $u_{(1)} \geq \dots \geq u_{(n)}$ give

$$\max_{\pi} M(\pi) = b + \sum_i a_{(i)}u_{(i)}, \quad (5)$$

$$\min_{\pi} M(\pi) = b + \sum_i a_{(i)}u_{(n+1-i)}. \quad (6)$$

No statistic invariant to token permutation can determine evidence assignment or the output margin.

Proof. An r -position set receives at most the r largest weights and at least the r smallest; assigning those weights to S attains the endpoints. Normalizing the bounded mass gives ρ_S . For the output range, suppose $u_p > u_q$ but $a_{\pi(p)} < a_{\pi(q)}$. Swapping those weights changes the margin by $(a_{\pi(q)} - a_{\pi(p)})(u_p - u_q) > 0$. Repeated exchanges align weights and utilities for the maximum and reverse them for the minimum, the classical rearrangement inequality (Hardy, Littlewood, and Pólya 1952). The extremal assignments preserve the complete weight multiset and therefore every permutation-invariant statistic. $\square$

Nonlinear downstream network

Fix an input and selected heads H at one layer. Let $x_{hj} = O_h v_{hj}$ and $z = \sum_{h \in H} \sum_j a_{hj} x_{hj}$. For target y and competitor c , let $M_c(z) = g_y(z) - g_c(z)$. A source-donor swap has

$$\delta_h = a_{hd_h} - a_{hs}, \quad \Delta z = \sum_{h \in H} \delta_h (x_{hs} - x_{hd_h}). \quad (7)$$

Define local utility $u_{hj}^{(c)} = \nabla M_c(z)^\top x_{hj}$.

Theorem S2 (Nonlinear routing-margin bound). *If ∇M_c is L_c -Lipschitz from z to $z + \Delta z$, then*

$$\left| \Delta M_c - \sum_{h \in H} \delta_h (u_{hs}^{(c)} - u_{hd_h}^{(c)}) \right| \leq \frac{L_c}{2} \|\Delta z\|_2^2. \quad (8)$$

Main-paper claim	Supplement evidence	Saved artifact family	Analysis entry point
Capacity and assignment are distinct	Theory details; controlled permutation and concentration controls	<code>factorial_grid</code> , paired permutation records	Concentration and assignment analysis
Assignment effects depend on downstream utility	Nonlinear bound, held-out utility calibration, and dose response	Controlled utility records, pretrained utility records, dose-response records	Controlled and pretrained utility analyses
The same swap exposes different pretrained regimes	Full family table, selector comparison, layer and head controls	Pretrained causal, selector, and circuit-selection records	Pretrained routing and selector analyses
The intervention is local and conditional	Arity gate, activation patching, exact Qwen rerun, endogenous-row comparison, and natural-QA ladder	Multi-evidence, patching, replication, endogenous, and natural-QA records	Robustness and boundary analyses

Table S1: Claim-to-artifact map. Paths are relative to `results/lengthgen/` in the anonymous bundle. All reported statistics can be reconstructed from saved artifacts without loading a language model.

Proof. The swap changes head h by exactly $\delta_h(x_{hs} - x_{hd_h})$. The fundamental theorem of calculus gives

$$\Delta M_c = \int_0^1 \nabla M_c(z + t\Delta z)^\top \Delta z dt. \quad (9)$$

Subtracting $\nabla M_c(z)^\top \Delta z$ and applying Lipschitzness bounds the remainder by $\int_0^1 L_c t \|\Delta z\|_2^2 dt = L_c \|\Delta z\|_2^2 / 2$ (Nesterov 2004). Expanding the first-order term gives Equation 8. $\square$

Corollary S3 (Multi-evidence routing). *Let S contain r evidence positions and pair every evidence position missing a top- r weight with one top- r non-evidence donor. For disjoint swap pairs P_h and $\delta_{hsd} = a_{hd} - a_{hs} \geq 0$,*

$$\Delta z = \sum_{h \in H} \sum_{(s,d) \in P_h} \delta_{hsd} (x_{hs} - x_{hd}), \quad (10)$$

and Equation 8 holds with the first-order sum extended over $(s, d) \in P_h$. Thus assigning top- r capacity to the evidence improves the margin whenever the summed evidence-donor utility gain exceeds the curvature remainder.

The decomposition and corollary are algebraic for arbitrary r . Their behavioral premise is established for one and two evidence positions in the current experiments. The three-evidence interval includes zero, and the four-evidence models miss the competence gate, so efficacy beyond $r = 2$ remains open.

The condition is sufficient, not necessary. Source-max can fail when evidence values have no greater downstream utility, patched heads cancel, or curvature dominates. Natural source mass alone is therefore not a universal circuit-selection rule.

Controlled Experiments

Tasks and training

Argmax retrieval presents key-value pairs and requests the value paired with the largest key. **Flag retrieval** presents values with exactly one marked position and requests the marked value. Both tasks are order-invariant and have one identifiable evidence position. **Reversed decimal addition**

is an order-dependent contrast. **Two-evidence modular sum** marks two digits and requests their sum modulo ten, so copying either digit is insufficient.

The primary model is a four-layer pre-norm decoder transformer with width 256, eight heads, and feed-forward width 1024. We train with AdamW, batch size 512, and warmup plus cosine decay on lengths one through five. Evaluation follows a geometric ladder to $50\times$ training length. The scale replication uses eight layers and width 512. A cell is interpreted only after training-length exact match reaches 0.8. Order-invariant cells use four seeds; the order-dependent contrast uses two. The preregistration fixes the competence gate, break-point rule, and variance-collapse check before intervention outcomes.

Measured quantities and interventions

At the answer query we record evidence mass, attention entropy, every attention norm, participation ratio, attention-output variance, and target margins. The variance intervention replaces $h \leftarrow h + z$ with $h \leftarrow h + \text{LN}(z)$ after attention, holding the downstream variance of z approximately constant. The concentration intervention applies length-scaled logit sharpening.

For the fixed-spectrum test, source-max swaps the source weight with the row maximum, source-min swaps it with the row minimum, and the distractor control swaps the largest and smallest non-source weights. All conditions preserve model parameters, prompts, transformed values, the complete sorted attention row, entropy, norms, maximum, and participation ratio. The control additionally preserves source mass.

Confirmatory inference. We define one family of three primary contrasts: controlled source-max minus distractor control, the capacity-by-assignment interaction, and natural utility-selected source-max minus matched control. Exact two-sided sign-flip tests operate on independent trained-model clusters for the controlled claims and calibration-seed clusters for natural QA. Holm adjustment controls family-wise error across the three tests. Hierarchical intervals and

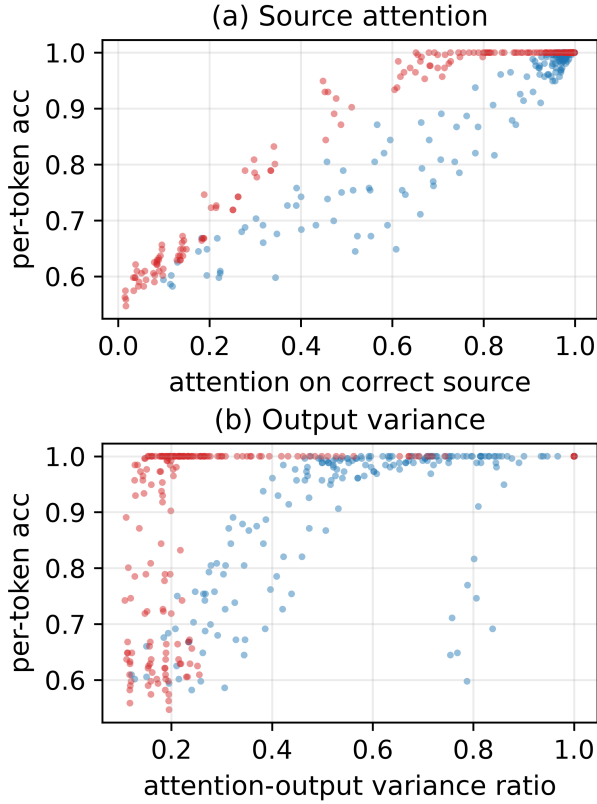


Figure S1: Controlled baselines pooled over lengths and seeds. Source attention has pooled Spearman/Kendall 0.854/0.713, compared with 0.445/0.338 for output variance; the ordering remains on non-ceiling points.

all other reported cells are descriptive and unadjusted unless stated otherwise.

Observational and variance-control results

Of the 384 aggregate controlled points, 151 (39.3%) lie at exact accuracy ceiling. Rank analyses remain separated despite that pileup: source attention has pooled Spearman 0.854 and Kendall $\tau_b = 0.713$, versus 0.445 and 0.338 for attention-output variance. On the 233 non-ceiling points, the corresponding values are 0.912/0.751 and 0.544/0.402. Mean within-cell Spearman is 0.947 for attention and 0.777 for variance. The variance intervention holds the post-intervention ratio near one, yet accuracy remains impaired (Table S2).

Capacity, assignment, utility, and dose

Across sixteen trained models, source-max raises accuracy by 0.153 (95% CI [0.086, 0.222]), source-min lowers it by 0.043 ([−0.058, −0.030]), and the distractor control changes it by 0.011 ([0.004, 0.018]). Source-max exceeds control in all 32 model-length comparisons. Maximum invariant error is below 10^{-6} . Increasing patched heads from one, two, four, to eight gives mean gains 0.007, 0.013, 0.056, and 0.153, monotone in 31 of 32 comparisons.

The confirmatory source-max minus control contrast averages +.142 across 16 trained-model clusters (95% cluster bootstrap [+0.080, +.206]). Its two-sided exact cluster sign-flip test gives $p = 3.05 \times 10^{-5}$ and Holm-adjusted $p = 9.16 \times 10^{-5}$ within the three-claim primary family.

At temperatures $\beta = 1, 2, 4$, mean maximum weight rises from 0.362 to 0.655 and the source-max minus source-min contrast rises from 0.196 to 0.346, an interaction of +0.151 ([0.094, 0.210]). The corresponding model-cluster estimate is +.150 ([+0.094, +.210]; exact $p = 6.10 \times 10^{-5}$; Holm-adjusted $p = 1.22 \times 10^{-4}$). Sharpening without reassignment changes accuracy by at most +0.009. Across 1,024 examples, the local utility term predicts finite margin changes with Pearson 0.917, Spearman 0.989, and 99.5% sign agreement. Active source utility gaps are positive in 89.2% of head-example pairs.

Directly forcing source mass provides a separate dose response: at $50\times$ training length, baseline accuracy is 0.59 and rises monotonically toward one as forced source mass approaches one. In this controlled task, the routed source value is sufficient; arbitrary source-max swaps need not satisfy the utility condition.

Two-evidence computation and arity boundary

Eight two-layer width-128 models reach perfect training competence on modular sum. At $5\times$ length, evidence-max raises exact match by 0.066, evidence-min lowers it by 0.291, and the matched control changes it by 0.023. Evidence-max exceeds control by 0.043 ([0.017, 0.072]).

The variable-evidence study does not establish effects for arbitrary r . Two- and three-evidence models reach training competence, but the three-evidence max-control interval includes zero. Four-evidence candidates fail the frozen competence gate: the largest eight-layer, width-512 candidate reaches 0.748 exact match against the required 0.8. No four-evidence intervention is interpreted.

Pretrained-Model Protocols and Results

Synthetic in-context recall

Prompts contain key-value pairs followed by a query key. The source is the value paired with that key. For each model, a calibration split selects a layer and K heads; this circuit is frozen across disjoint evaluation examples and lengths. We measure untouched accuracy, target-versus-competitor margin, and every intervention invariant on paired examples. The family probe uses Pythia-1.4B, Qwen2.5-1.5B, and Gemma-2-2B; causal tests additionally use Qwen2.5-7B NF4 and SmolLM2-1.7B.

Utility-conditioned circuit selection

For every candidate head, the calibration split evaluates transferred source mass times the local source-donor utility gap. Selecting the layer with the largest top-four sum produces a positive SmolLM2 effect over three seeds: at five pairs, source-max minus control is +0.702 margin ([0.529, 0.858]) and +0.052 accuracy, while the source-mass selector changes margin by −0.021 ([−0.057, 0.014]). The paired utility-minus-mass effect is +0.723 ([0.544, 0.882]).

Task	PE, intervention	1×	3×	10×	50×	Var. (pre, post)	Effect at 50×
Argmax	NoPE, off	1.00	.98	.83	.59	.33, .33	n/a
	NoPE, on	1.00	.97	.80	.58	.20, .99	−.013
	RoPE, off	1.00	.95	.77	.60	.27, .27	n/a
	RoPE, on	1.00	.96	.67	.57	.45, 1.00	−.033
Flag	NoPE, off	1.00	.94	.67	.60	.47, .47	n/a
	NoPE, on	1.00	.88	.62	.52	.49, 1.00	−.081
	RoPE, off	1.00	.99	.65	.56	.48, .48	n/a
	RoPE, on	1.00	.96	.53	.50	.43, 1.00	−.061

Table S2: Controlled retrieval accuracy and variance stabilization. The variance columns report the long-length ratio before and after normalization. Behavior remains impaired after the statistic is repaired.

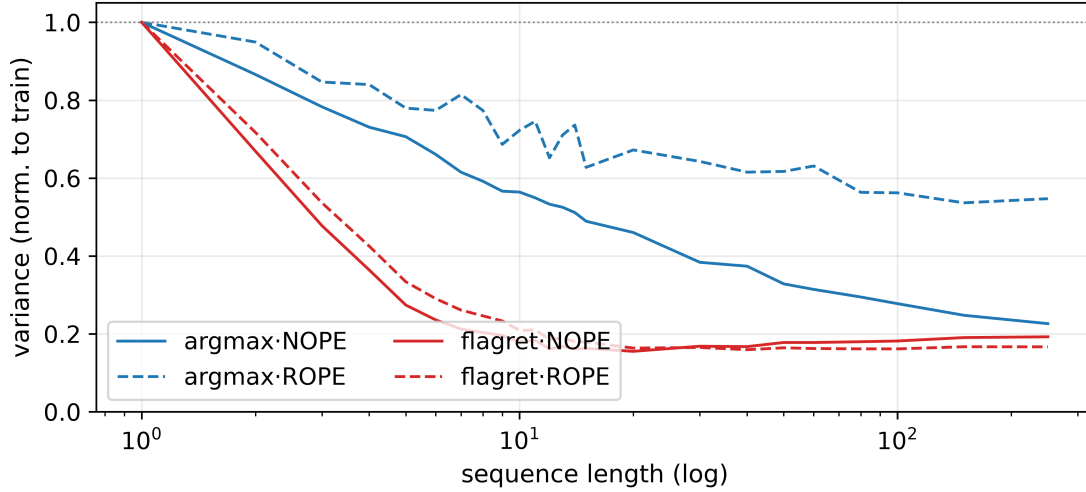


Figure S2: Baseline attention-output variance collapses with length on every controlled task.

Length	Natural	Evid.-max	Evid.-min	Control	Max-ctrl.
2×	.952	.971	.393	.964	+.007
5×	.484	.550	.194	.507	+.043

Table S3: Exact match on two-evidence modular sum, averaged over eight competent models.

The ten-seed equal-budget ablation compares source mass, transfer mass, utility gap, predicted utility gain, source gradient, gradient magnitude, and deterministic random circuits. Source gradient ranks first at +0.750 ([0.682, 0.819]), utility gain second at +0.600 ([0.482, 0.712]), utility gap third at +0.296, and source mass is −0.040. Across Qwen, Pythia, and SmolLM2, no single selector wins every model; output-conditioned selectors nevertheless beat task-agnostic source mass.

Model	Positive cells	Mean Spearman	Sign agreement
Qwen2.5-1.5B	6/6	.866 [.832,.898]	.805
Pythia-1.4B	6/6	.544 [.500,.586]	.807
Gemma-2-2B	6/6	.554 [.534,.582]	.714

Table S6: Three-seed pretrained utility replication. Brackets give the range of seedwise source-max Spearman correlations.

Held-out utility calibration and intervention displacement. The utility analysis uses circuits selected by source mass on calibration examples, then evaluates the local first-order term on disjoint rows. It therefore does not select a circuit by the same per-row utility values used in the prediction check. Table S7 reports calibration as well as rank agreement. The first-order term is informative but imperfect, particularly for the largest Qwen effects, so we use it only as a local ranking signal. We also retain rows where every selected head already assigns the source its maximum weight. These rows have zero source-max displacement and attenuate the aggregate effect.

Across 8,192 controlled head-example pairs, median transferred mass is .025 and the 90th percentile is .930. Only 86 of 1,024 complete circuits are entirely vacuous.

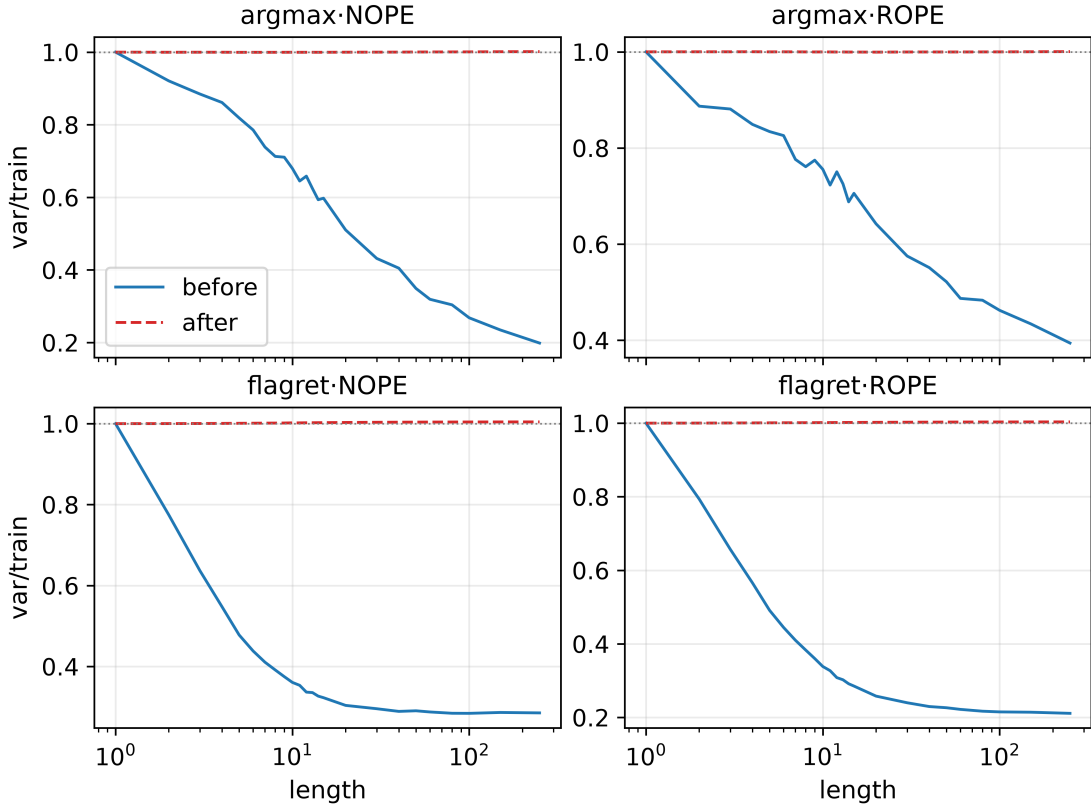


Figure S3: The variance intervention works mechanically: raw variance still collapses while normalized variance remains near its training-length scale.

Mean exact margin change is $+3.891$ over all circuits and $+4.248$ ($[+3.863, +4.640]$) over the 938 active circuits. Binning examples by total transfer gives means of $+.081$, $+.277$, $+2.436$, and $+12.769$ from the lowest to highest quartile. Vacuous rows therefore attenuate the aggregate rather than create it.

Metric	Qwen2.5-1.5B	Pythia-1.4B	Gemma-2-2B
Evaluation rows	384	384	384
Pearson	.827	.670	.738
Slope	.735	.885	.947
Intercept	+.218	+.184	+.001
MAE	.378	.724	.112
All selected heads source-max	7.8%	4.2%	21.9%
Median transferred mass	.783	.346	.312

Table S7: Held-out calibration and displacement results for the pretrained utility check. Slope and intercept fit exact margin change from the first-order prediction. All rows are retained, including zero-displacement source-max cases. The reproducible aggregation reads the saved per-example artifacts.

Activation-patching baseline. We compare fixed-spectrum swaps with standard clean-to-corrupt activation patching on Qwen2.5-1.5B. For each held-out row, source-value corruption replaces the labeled value while preserving

prompt length and positions. We then patch the selected per-head attention outputs from the clean run into the corrupted run. Circuit selection uses a disjoint 64-example split; each seed evaluates 128 paired rows. The fixed-spectrum condition acts on the unchanged clean prompt, so its assignment effect and the activation-patch restoration effect are different estimands and are not subtracted.

Exact Qwen protocol replication. Five new seeds repeat the headline protocol with the same bfloat16 precision, four-head calibration rule, 5/20/80/160-pair ladder, 128 examples per length, and source-max minus distractor-control contrast. Every new seed is positive, but the effect is much smaller after the selected circuit moves from layer 19 to layer 14. The five new means average $+.042$; their exact two-sided test has the five-seed minimum $p = .0625$. Combining the preregistered new seeds with the known seed 0 gives $+.349$ ($[+.037, +.982]$, $p = .03125$).

Endogenous assignment bridge. For each untouched Qwen prompt, the source pair remains in the same slot while distractor order varies. We match the pair of naturally produced attention rows with minimum sorted-spectrum distance subject to a source mass gap of at least $.01$. The paired margin effect is $+.026$ ($[-.017, +.070]$, $p = .25$). A within-instance regression that controls maximum weight and entropy gives a positive source-mass coefficient in every seed,

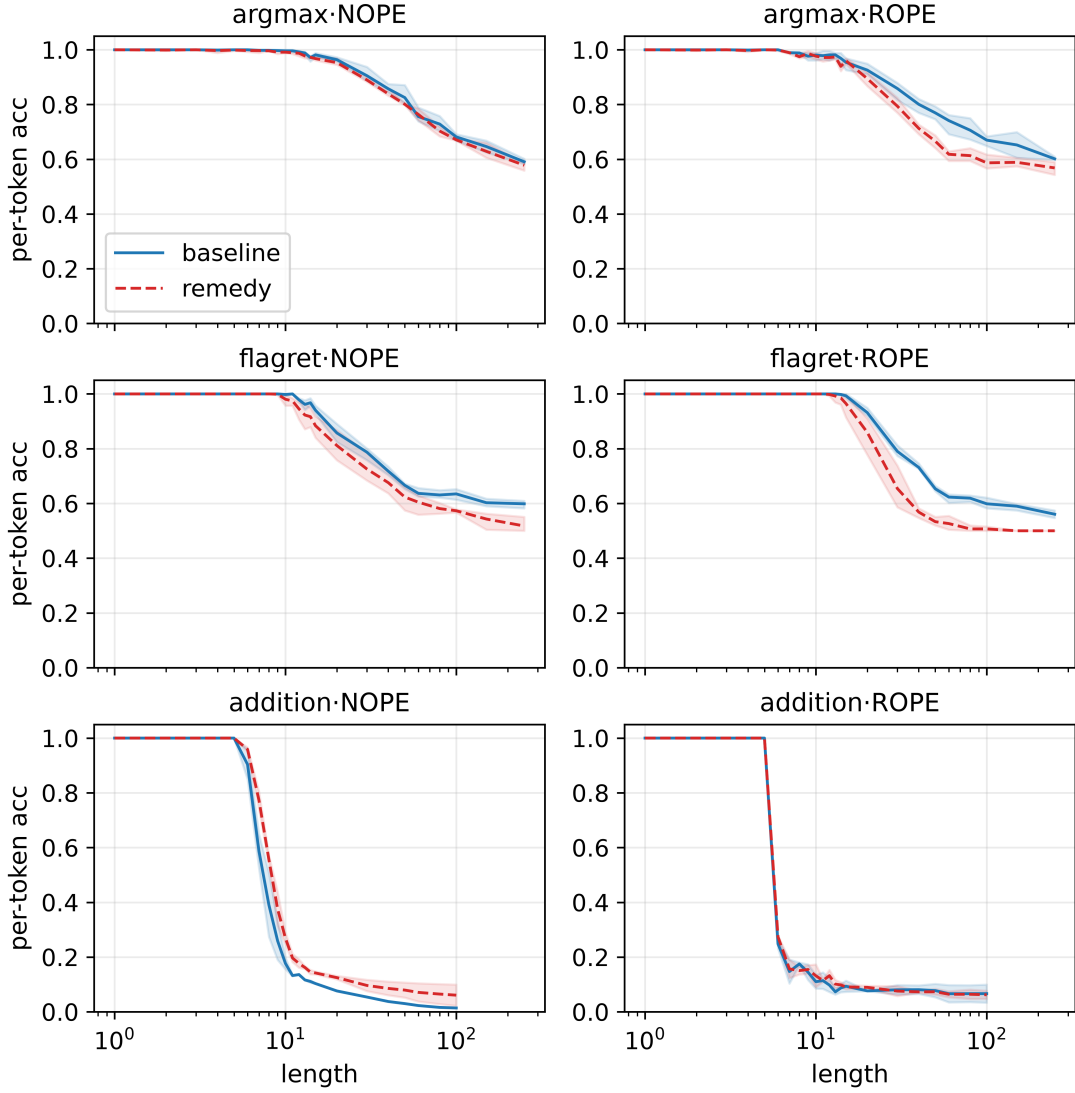


Figure S4: Accuracy across the full length ladder. The variance intervention lies at or below baseline in all retrieval cells.

averaging $+1.250$ ($p = .03125$). These endogenous rows support an association with source mass; the stricter matched comparison remains inconclusive and does not identify an endogenous causal mechanism.

Continuous interpolation supplies a dose control. At strengths 0, .25, .5, .75, 1, utility-selected source-max minus matched distractor interpolation changes margin by 0.000, 0.446, 0.817, 1.128, and 1.417. Every one of five seedwise paths is nondecreasing.

Natural multiple-choice QA

The natural test uses four-choice questions with a gold passage and unrelated passages. An example is eligible only if the model answers correctly with full context and the gold passage alone, but incorrectly without context. This filtering defines a model-specific context-dependent subset and is not intended to estimate average performance over the

full dataset. Calibration and evaluation examples are disjoint. Across the 1,536 evaluation-row and selector combinations, define ϵ_i as mean selected-head source-max L_1 displacement minus mean selected-head matched-control displacement. The median $|\epsilon_i|$ is 2.59×10^{-4} , its 95th percentile is 2.32×10^{-3} , and its maximum is 4.54×10^{-2} . On the locked 64-calibration/128-evaluation design over six seeds, utility-selected source-max exceeds control by $+0.378$ margin ($[0.253, 0.520]$). Source-mass selection is harmful at -0.090 ($[-0.141, -0.049]$). An eight-token greedy check changes neither first-answer accuracy nor repetition relative to control. All six utility-selected seed means are positive; the exact two-sided sign-flip test gives $p = .03125$, which remains .03125 after Holm adjustment across the three confirmatory tests.

For the nested length ladder, the selected circuit is frozen at four passages before appending unrelated passages. Main

Model	Pairs	Base	Δ accuracy			Δ margin	
			Acc.	Source-max	Source-min	Control	Source-max
Qwen-1.5B	5	.539	+ .102	- .258	- .016	+1.177	-1.406
	20	.398	+ .188	- .117	+ .023	+2.268	-1.191
	80	.070	+ .109	- .023	+ .000	+2.527	- .285
	160	.008	+ .062	+ .000	+ .000	+2.204	- .148
Qwen-7B NF4	5	.448	+ .021	- .135	+ .010	+ .325	- .988
	20	.271	+ .135	- .104	+ .010	+ .891	-1.185
	80	.010	+ .010	- .010	+ .000	+ .654	- .546
Pythia-1.4B	5	.359	+ .109	- .250	+ .039	+ .878	-2.480
	20	.039	+ .039	- .039	- .016	+1.195	- .622
	80	.000	+ .000	+ .000	+ .000	+ .405	+ .034
Gemma-2-2B	5	.758	+ .008	- .094	+ .016	+ .041	-1.680
	20	.711	+ .016	- .133	- .008	+ .020	-1.341
	80	.742	+ .000	- .117	+ .000	+ .058	-1.096
SmolLM2-1.7B	5	.945	+ .008	- .008	- .008	- .013	- .133
	20	.516	- .016	+ .000	+ .008	- .410	- .190
	80	.023	+ .000	+ .000	+ .000	- .083	- .012

Table S4: Source-mass-selected intervention results for seed 0, with 128 paired held-out examples in each model-length cell. Qwen and Pythia show positive source-max effects; Gemma is saturated; SmolLM2 is the preregistered negative replication. Main Figure 5 uses the common 5-, 20-, and 80-pair cells only, so every curve has identical seed, sample count, and length support.

Figure 6 compares seed 0 for both models, using the same 128 paired questions in every model-length cell. Table S12 separately reports the broader available-seed aggregation: Qwen seeds 0 and 2 and SmolLM2 seed 0, with 128 examples per seed and passage count. Qwen loses performance while source mass rises and rescue weakens. SmolLM2 loses performance while source mass falls and rescue does not increase. The two models therefore do not share a source-mass-loss or increasing-rescue account of natural-context degradation.

Seed	K	Circuit	Layer	Max-control ΔM	Paired 95% interval
0	2	selected	14	+0.641	[.311,1.018]
0	4	selected	14	+1.430	[.919,2.003]
0	8	selected	14	+1.418	[.885,2.013]
0	4	adjacent -1	13	+0.067	[-.091,.231]
0	4	adjacent +1	15	-.355	[-.568,-.181]
0	4	random layer	11	-.093	[-.227,.029]
1	2	selected	14	+0.459	[.167,.801]
1	4	selected	14	+0.981	[.507,1.539]
1	8	selected	14	+1.069	[.584,1.598]
1	4	adjacent -1	13	+0.049	[-.085,.189]
1	4	adjacent +1	15	-.277	[-.395,-.168]
1	4	random layer	21	-.190	[-.326,-.059]
2	2	selected	14	+0.281	[.025,.550]
2	4	selected	14	+0.749	[.356,1.190]
2	8	selected	14	+0.835	[.427,1.288]
2	4	adjacent -1	13	+0.326	[.146,.548]
2	4	adjacent +1	15	+0.239	[.065,.457]
2	4	random layer	7	+0.106	[-.007,.223]

Table S5: Qwen layer, head-budget, and calibration-seed robustness results. All nine selected-circuit intervals exclude zero; none of the three random-layer intervals does.

Metric	Seed 0	Seed 1	Seed 2
Layer	14	14	14
Fixed-spectrum ΔM	+0.012 [.001,.024]	+0.030 [-.006,.085]	+0.021 [-.004,.051]
Corruption ΔM	-7.726 [-8.185,-7.273]	-7.486 [-7.878,-7.105]	-7.779 [-8.242,-7.333]
Patch rescue	+0.369 [.295,.452]	+0.275 [.205,.351]	+0.280 [.209,.358]
Median damage recovered	3.3%	2.6%	2.5%
Invariant error	2.4×10^{-7}	2.4×10^{-7}	2.4×10^{-7}

Table S8: Fixed-spectrum reassignment and activation patching on independently calibrated Qwen circuits and held-out rows. Brackets are paired 95% intervals within each seed. Activation patching partially restores information removed by source-value corruption; fixed-spectrum reassignment tests assignment without changing the prompt. The fixed-spectrum effects are smaller than the primary Qwen result because these rows independently calibrate a four-head float32 circuit at five pairs for each seed (layer 14) and report source-max minus the untouched baseline on 128 rows. Table 1 uses one seed-0 bfloat16 circuit calibrated at five pairs (layer 19), averages 128 rows at each of 5, 20, 80, and 160 pairs, and reports source-max minus distractor control ($n = 512$).

Seed	Layer	Mean ΔM	Mean $\Delta \text{acc.}$
0	19	+1.882	+0.113
1	14	+0.041	+0.002
2	14	+0.036	+0.012
3	14	+0.053	+0.018
4	14	+0.039	+0.002
5	14	+0.044	+0.018

Table S9: Exact matched Qwen rerun. Seed 0 is the previously observed result; seeds 1–5 were registered before their outcomes. Direction replicates, while the original magnitude does not.

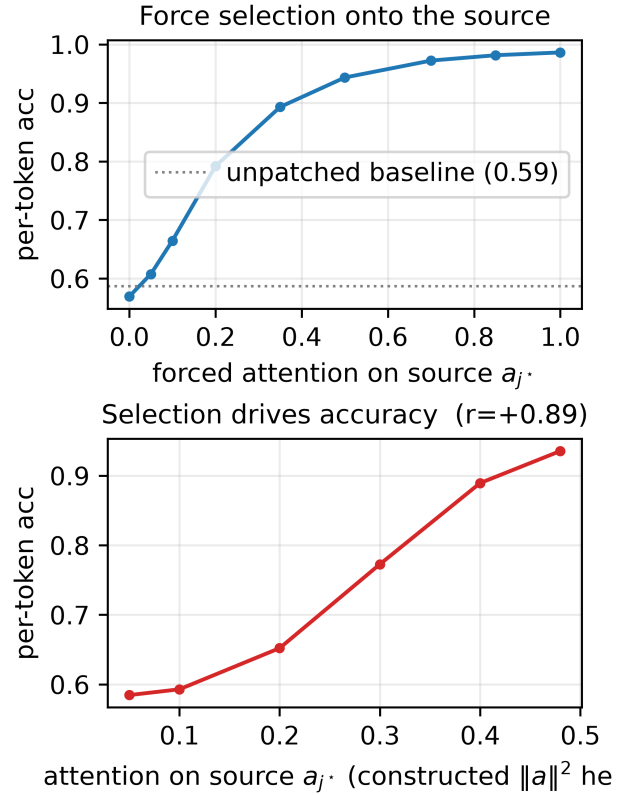


Figure S5: Direct manipulation of source attention. Accuracy rises monotonically with forced source mass.

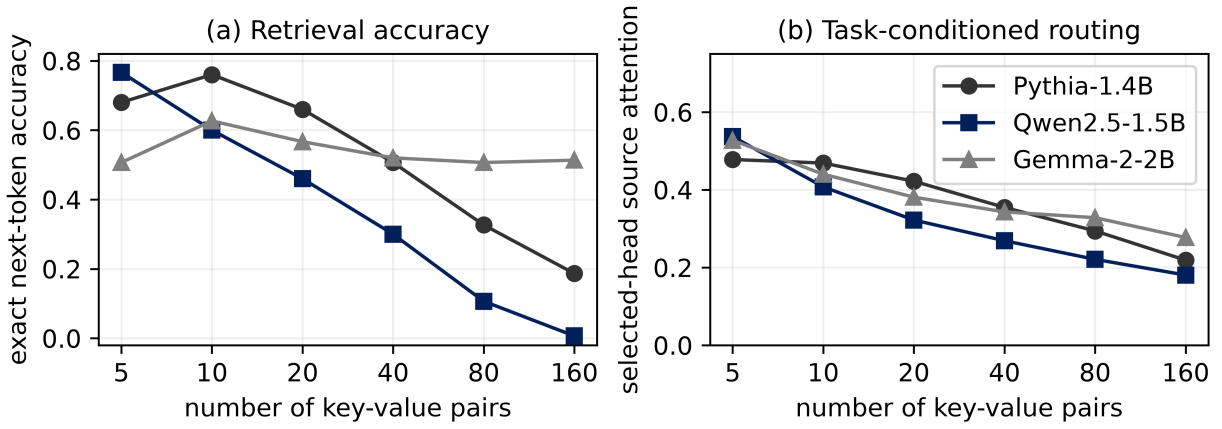


Figure S6: Pretrained family sweep. Pythia and Qwen lose accuracy with length, while Gemma remains competent despite declining selected-head source attention.

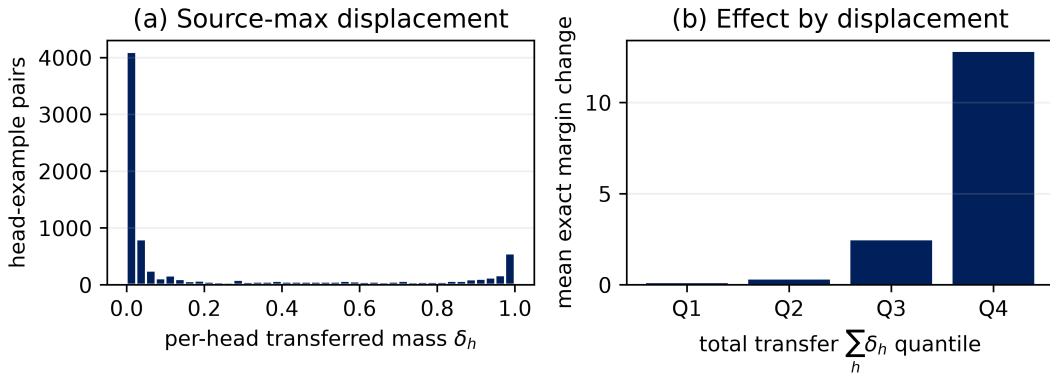


Figure S7: Controlled source-max displacement. Left: the per-head transferred-mass distribution. Right: exact margin change increases with total circuit transfer. Individual swaps are vacuous for 18.5% of head-example pairs, while 8.4% of complete eight-head circuits are entirely vacuous.

Seed	Matched ΔM	Positive	Source coef.	Spectrum L_1
0	-.002	48.4%	+.939	.1226
1	-.020	44.9%	+1.087	.1244
2	+.069	51.6%	+1.360	.1210
3	+.075	53.5%	+1.442	.1230
4	+.038	48.4%	+1.857	.1202
5	-.001	46.9%	+.815	.1216

Table S10: On-manifold distractor-order comparison across six calibration and evaluation seeds, with 256 eligible pairs per seed. Coefficients are within-instance source-mass slopes controlling maximum weight and entropy.

Model	Top selector	Effect	Utility rank
Pythia-1.4B	utility gain	+.519	1
SmolLM2-1.7B	utility gain	+1.253	1
Qwen2.5-1.5B	utility gap	+1.493	2

Table S11: Equal-budget cross-family selector comparison. No selector wins all three families; output-conditioned selectors lead each family.

Metric	Qwen2.5-1.5B	SmolLM2-1.7B
Seeds	0, 2	0
n per seed	128	128
Δ source mass	+.000955 [+ .000141, +.002033]	-.0000736 [-.0000989, -.0000519]
Δ margin	-1.036 [-1.397, -.698]	-.193 [-.320, -.066]
Δ accuracy	-.0469 [-.0742, -.0234]	-.1016 [-.1563, -.0547]
Δ rescue	-.179 [-.352, -.005]	+.018 [-.027, +.066]

Table S12: Natural-QA 32-minus-4-passage changes with hierarchical 95% intervals. This robustness table uses all passing seeds available for each model and is separate from the seed-0 matched comparison in Main Figure 6.

Alternative Explanations and Controls

Alternative	Matched control and quantitative result	Conclusion
Concentration changed	Exact invariant checks give sorted-spectrum error 0 in controlled models and $\leq 4.77 \times 10^{-7}$ in pretrained runs.	Spectrum invariant
Any perturbation helps	On the same rows, distractor-only control changes accuracy by +.011 [.004,.018] versus source-max +.153 [.086,.222] .	Control is insufficient
Total source mass suffices	In the two-evidence mass-preserving test, evidence-max exceeds the same-mass control by +.043 [.017,.072] exact match.	Not supported
Output variance is causal	Variance normalization returns the ratio to .99–1.00, while $50 \times \Delta\text{accuracy}$ remains -.013 to -.081 .	Not supported
Any layer works	Frozen selected Qwen cells average +.874 ; random layers average -.059.	Not supported
Output-independent selection suffices	Equal-budget ablation gives source gradient +.750 , utility gain +.600 , and source mass -.040 .	Not supported

Table S13: Alternative explanations, matched controls, and their quantitative outcomes.

Table S13, continued.

Alternative	Matched control and quantitative result	Conclusion
Calibration leakage drives the gain	Disjoint calibration and evaluation yield +.378 [.253,.520] held-out natural-QA margin over control across six seeds.	Addressed by split
One source-mass trajectory explains natural length failure	Qwen Δm_S is +.00095 while margin falls -1.036 ; SmolLM2 Δm_S is -.00007 with flat rescue.	Not shared across models
The result extends to arbitrary evidence arity	Registered effects are pair +.043 [.017,.072] , triple -.001 [-.009,.008], while quadruple misses competence.	Unresolved beyond two
Headline Qwen direction is seed-specific	Five new seeds are all positive with mean +.042 ; the original +1.882 is atypically large.	Direction replicates; magnitude unstable
Off-manifold swap reflects endogenous rows	Untouched variants give source coefficient +1.250 ($p = .03125$) and matched effect +.026 [-.017,+.070].	Partial bridge; causal gap remains

Scope and Boundaries

The equals-sign second prompt format fails its competence gate in Qwen; the blinded arrow-format pilot also fails full-run competence despite directional margin signs. These runs are excluded from causal replications. Pythia and Qwen are floor-limited at their longest contexts, so margin is the interpretable tail outcome. Qwen-7B is quantized. Gemma supplies a high-competence saturation boundary.

The fixed-spectrum theorem remains applicable to the natural-QA length result because it conditions effects on transferred mass, value utility, and curvature. The length ladder does not support the stronger claim that natural length degradation must involve source-mass loss or become increasingly rescuable by the same frozen circuit.

Reproducibility

Code and artifacts. The anonymous supplementary bundle contains the paper-specific training, preprocessing, intervention, analysis, and figure code; configurations and random seeds; prompts; paired per-example outputs; competence-gate failures; invariant checks; and preregistrations. The saved artifacts reproduce reported statistics and figures without rerunning a model, while the runners reproduce the experiments from public pretrained checkpoints or newly trained controlled models. `ARTIFACT_INDEX.md` maps claims to files, `DATA_DICTIONARY.md` defines common fields and units, and `REPRODUCE.md` gives CPU-only analysis, test, figure, and optional GPU commands.

Data and models. The controlled tasks are generated from the included task generator and therefore require no external dataset. Natural MCQA samples are constructed from the public SQuAD training split (Rajpurkar et al. 2016); the sampling seed and eligibility rule are included in the bundle. Pretrained probes use publicly released Qwen, Pythia, Gemma, and SmolLM2 checkpoints, with exact identifiers,

tokenizer settings, and precision recorded in the result artifacts. Later pretrained experiments used a Linux Google Colab runtime with one NVIDIA T4 GPU (16 GB), PyTorch, and Hugging Face Transformers. The earlier controlled artifacts retain configurations and seeds but not the accelerator model or exact package-version snapshot; we therefore mark infrastructure reproducibility partial in the checklist. A public code and artifact archive will be released on acceptance.

Integrity and anonymity. `MANIFEST.sha256` records the path, byte size, and SHA-256 digest of every bundled file except the manifest itself. Running `python scripts/verify_package.py` checks required assets, parses every JSON artifact, scans text files for local user paths and credential-shaped strings, and verifies the manifest. The upload archive omits Git history, model caches, compiled paper PDFs, and LaTeX build products.

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